Name: _____ Period: _____

Review and Reassessment Workshop

Tier: All tiers **In class:** Targeted practice by tier; TIPERs review

About this page

- Use your target tracker to choose your section. Start with the lowest tier that is not yet secure. A higher tier counts only when the tiers below it are secure.

Key ideas

- Foundations: problems 1–3. Core Skills: problems 4–7. Target: problems 8–10. Everyone: problems 11–14 (graphs, a paragraph argument, and designing an experiment).
- Check your work against the key your teacher posts, then redo one missed problem from the matching section.

1. Foundations. Find the resultant of 7.0 km east and 24 km north.

2. Foundations. Resolve 40 m/s at 65° above the horizontal into components.

3. Foundations. A ball is thrown horizontally off a cliff. Describe what happens to its horizontal velocity, vertical velocity, and acceleration during the fall.

4. Core Skills. Add 12 m east and 9.0 m at 70° N of E.

5. Core Skills. Ivy bats a toy ball off a 2.2 m shelf, and it leaves moving horizontally at 4.0 m/s. Find the time to land and where it lands.

6. Core Skills. For the ball in problem 5, find v_x and v_y at $t = 0.30$ s.

7. Core Skills. A kayak heads straight across a 50 m river at 2.0 m/s with a 0.80 m/s current. Find the time, drift, and velocity relative to the bank.

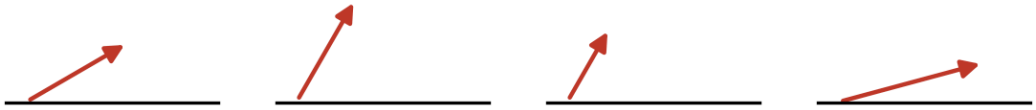
8. Target. A ball is kicked at 30 m/s at 40° on level ground. Find its time of flight, maximum height, and range.

9. Target. Using the cannon in Section 2.11, how would the landing spot change if the cliff were taller? Explain with the columns, not a formula.

10. Target. A plane heads due east at 160 m/s in a 30 m/s wind toward the north. Find its ground velocity. Then find the heading it needs to travel due east.

11. Ranking (everyone). Four balls are launched from level ground. Rank them, greatest first, by a) time of flight and b) range. Explain one ranking.

a) _____ b) _____



A: 20 m/s at 30° B: 20 m/s at 60° C: 10 m/s at 60° D: 30 m/s at 15°

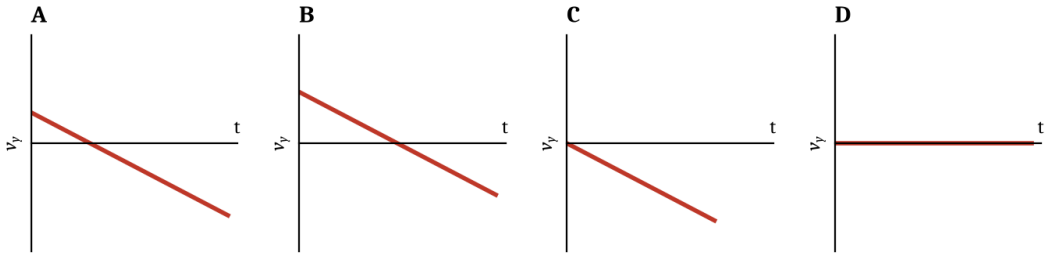
12. Graph matching (everyone). Each graph shows v_y vs. t (up positive) for one motion, from launch to landing. Match each motion to a graph, then say what all the slanted graphs have in common and why.

1) a marble rolling across a level tabletop 2) a ball rolling off a table

3) a ball kicked from level ground that lands at the same height

4) a ball thrown upward at an angle from a cliff that lands below the cliff

Matches: 1 ___ 2 ___ 3 ___ 4 ___



13. Paragraph argument (everyone). Two balls are launched at 26 m/s from level ground, one at 30° and one at 60°. A classmate claims: “They land at the same spot, so they must be in the air for the same time.” Do you agree? Write a paragraph argument: a claim, evidence (a physics principle or equation), and reasoning that links them, in 3–5 complete sentences.

14. Design an experiment (everyone). Design an experiment to measure g using a ramp, a lab table, carbon paper, a meter stick, and a phone that records slow-motion video. Your plan must:

a) name every quantity you will measure and the equipment for each;

b) state the independent, dependent, and controlled variables;

c) describe how you will repeat trials;

d) say what you will graph to get a straight line and how you will find g from the slope.

15. Foundations. Mrs. Kraus does a running jump, leaving the ground at 12 m/s, 50° above the horizontal. Resolve her launch velocity into components.

16. Core Skills. A marble rolls off a 1.0 m table at 1.8 m/s. Where does it land?

17. Target. A ball is thrown from a 10 m roof at 12 m/s, 35° above the horizontal. How far from the building does it land?

18. Challenge (everyone). Mr. Lehman rides a motorcycle up a ramp that launches him at 20° above the horizontal. He must clear a 15 m gap and land on a platform 3.0 m lower than the end of the ramp.

a) Without finding t first, combine the x and y equations to find the minimum launch speed.

b) To be safe he launches 10% faster than that. How far past the near edge of the platform does he land?